The Edge Negotiator

Trustworthy On-Device Small-Language-Model Control of Urban Traffic Signals with a Provable Accident Audit

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Abstract

[TODO: Write last. Structure: (1) problem — can a 4 GB-class on-device SLM served by Microsoft Foundry Local control real London junctions safely while producing a legally grounded accident audit; (2) method — MaxPressure-shielded SLM control in SUMO on measured London topologies, frontier-distilled QLoRA fine-tuning, Ed25519 hash-chained audit with UK-law fault grounding; (3) headline results — offline distillation ceiling 97–99% at 0.6B (scale saturation), closed-loop delay decoupling, authorship conversion 0.46→0.12 override, MaxPressure reversal on real topologies, trust-coefficient battery; (4) contribution claims.]

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Chapter 1

Introduction

1.1 Motivation

Traffic signals are the most consequential actuators most cities own, and the research frontier has begun handing them to large language models: GPT-4-distilled controllers [19], RL-tuned reasoning models [31], and cooperative LLM agents [9] now populate the literature. Yet every one of these controllers is evaluated against a cloud endpoint, while the junction itself is an edge environment: a signal cabinet has no guaranteed backhaul, a hard memory envelope, and — uniquely among LLM application domains — a statutory duty to account for itself when a collision occurs. Between the literature and a deployable junction sit two unanswered questions: whether a language model small enough to live in the cabinet can control real junctions competently, and whether its decisions can be bound into a record an investigator can trust. This dissertation answers both together, because they are not separable: an accountability record is only meaningful if the component it certifies actually authored the behaviour.

The work is conducted with and for a Microsoft-supervised context, which fixes the serving stack as a hard constraint: Microsoft Foundry Local on a 6 GB consumer GPU, capping the model ladder at roughly 4B parameters. I treat the constraint as the research object rather than an inconvenience — the question is what *this* deployable envelope can do, not what a datacentre could.

1.2 Research questions

RQ1 Can an on-device SLM (≤4 GB, Foundry Local, RTX 2060-class), behind a deterministic safety shield, match or beat classical baselines (fixed-time, MaxPressure) on measured London topologies in SUMO?

RQ2 What does distilling a frontier teacher into the SLM buy, offline and closed-loop — and at what model scale does the benefit saturate?

RQ3 Can every actuation be bound into a tamper-evident audit that, after a simulated collision, yields a UK-law-grounded evidence pack rather than an unaccountable log?

RQ4 Does a trust-coefficient mechanism over neighbour reports withstand honest, blatant, and stealth misreporting?

1.3 Contributions

1. **The first documented end-to-end chain** from QLoRA fine-tuning through int4 ONNX compilation to Foundry Local serving of a custom model, with bit-identical CPU/GPU decision equivalence and four documented serving-reliability findings with mitigations (§4.4, §6.8).
2. **A scale-saturation result**: frontier distillation lifts a 0.6B student to the teacher-agreement ceiling (97–99%), and a 3.8B student lands on statistically identical numbers — capability is not the bottleneck at this task, the distillation ceiling is (§6.3).
3. **The authorship finding**, the dissertation's headline: offline accuracy decouples from closed-loop delay (neither student significantly beats the fixed-time floor at n=30), but fine-tuning cuts shield override from 0.46–0.60 to 0.11–0.18 and raises the SLM's authored share of served decisions to 0.90 — converting the system from shield-driven to model-authored control, which is the property a meaningful audit requires (§6.5).

4. **A scope-bounded MaxPressure reversal**: on every measured London topology in this study, MaxPressure is worse than naive fixed-time — it dominates only the synthetic grids the literature tests on — with a mechanism grounded in the finite-capacity and load assumptions of the underlying theorems (§6.1).

5. **A proven audit-to-law pipeline**: seven collision scenarios reconstructed from signed, hash-chained records, with tampering caught and the governing UK rule inferred deterministically from verified facts, emitting cited evidence packs rather than verdicts (§6.6).

6. **An honestly-scoped trust mechanism**: asymmetric, discount-only trust that contains blatant misreporting but is evaded by a calibrated stealth attacker — reported as a limitation, not defended away (§6.7).

1.4 Scope and thesis statement

The locked thesis question of this project is exploratory: *is an on-device Foundry-Local SLM traffic controller that matches or beats classical control on real London topology, while producing a provable accident audit, possible at all — and at what model scale and configuration?* It is a feasibility and scale-threshold study. Everything runs in SUMO; three real London topologies plus synthetic grids; demand magnitudes are measured, their realisations synthetic; comparators are classical (fixed-time, MaxPressure), not trained deep-RL systems, because the contribution is on-device feasibility plus auditability, not a new state of the art in delay. Limitations are reported as findings throughout — the evaluative stance is that a negative with a mechanism is worth more than an unexamined positive.

1.5 Dissertation structure

Chapter 2 positions the work against the classical, learning-based, and LLM signal-control literatures and the audit and shielding theory it builds on. Chapter 3 presents the architecture and defines the authorship metrics. Chapter 4 records the engineering, including the serving chain. Chapter 5 fixes metrics, statistics, and leakage controls. Chapter 6 reports the evidence in the order it was built. Chapter 7 answers the research questions, analyses the delay-decoupling mechanism, and states threats to validity. Chapter 8 concludes.

Chapter 2

Background and related work

2.1 Classical signal control and its guarantees

[evidence: ANALYSIS.md \S 3 (backpressure cluster)]

Deployed urban signal control still rests on two families. Fixed-time control serves a pre-computed cycle whose stage splits are engineered offline from historical counts; it optimises nothing online, but it is deterministic, starvation-free by construction, and indifferent to sensing noise. Its adaptive rival, MaxPressure [30], is the field's canonical strong baseline: each intersection independently serves the phase with the greatest "pressure" (upstream minus downstream queue lengths, weighted by turn ratios), and Varaiya proved this decentralised rule throughput-optimal — it stabilises every stabilisable demand. That guarantee, however, is bought with assumptions, and a critical reading of the backpressure literature shows that in central London each assumption fails structurally rather than marginally.

First, the optimality proof assumes unbounded queue storage. Gregoire et al. [15] prove unconditionally — for any topology — that linear-pressure control of exactly Varaiya's form ceases to be work-conserving once queue capacities are finite, and they construct explicit deadlocks; their repair, capacity-normalised pressure, is *not* part of standard MaxPressure. Central London's short inter-junction links mean small capacities everywhere, so the theorem's failure condition is standing, not exceptional. Second, the only stability result for a deployable (non-oracle) backpressure controller holds under heavy load only: Zaidi et al. [35] require every queue to exceed saturation flow, concede that sub-saturation queues "can unstabilize the queuing network", and describe stability outside heavy load as "still a challenging problem". Measured diurnal demand sits below that threshold for most of the day, precisely the regime in which the pressure signal chases arrival noise while a fixed cycle holds steady. Third, every favourable empirical figure in this lineage is a grid figure — roughly 90% of oracle backpressure on a uniform 21×21 grid, falling to about 80% under heterogeneous parameters — accompanied by the authors' own caveat that grid results cannot be extended to arbitrary networks. Fourth, junctions with give-way movements, gap acceptance, or gyratory circulation fall outside the model domain entirely: service in these theorems is binary and phase-determined, so no guarantee applies at all.

The field's own tables corroborate the predicted direction. AttendLight [25] reports MaxPressure *losing* to naive fixed-time by 5–19% on its least regular topologies (T-junction configurations) while winning on regular four-way grids — degradation strongest exactly where geometry departs from the grid. Against this stands a mandatory scope boundary: at least six independent studies [9, 10, 13, 19, 31, 33] show MaxPressure beating fixed-time by 7–58% on CityFlow grid-like networks, including nominally "real" Manhattan, Jinan, and Hangzhou topologies — but always under synthetic or replayed-uniform demand. I therefore claim the reversal reported in §6.1 only for the conjunction of real irregular geometry, measured demand, and SUMO dynamics; the same MaxPressure implementation behaves conventionally on my own synthetic grids, which pre-empts the objection that the baseline is simply broken. The gap this dissertation fills is that no prior work empirically tests MaxPressure against fixed-time on measured UK geometry under measured demand, still less connects the outcome to the theorems above.

2.2 Learning-based signal control

Deep reinforcement learning dominates the academic literature that classical control left behind. FRAP [36] introduced phase-competition invariances; CoLight [33] added graph-attention coordination across intersections; AttendLight [25] pursued a single attention-based policy transferable across intersection layouts; Wei et al. [32] survey the wider lineage. The results are genuinely strong — on the benchmarks the community chose. Those benchmarks are almost uniformly CityFlow grids with synthetic or replayed demand, and reporting is typically single-run, so the literature's headline numbers carry no variance estimates. Two further limits motivated the field's turn to language models: trained policies are opaque, offering an operator no account of *why* a phase was served, and the sim-to-real gap is largely unexamined because the simulators and the demand models are shared across papers — an evaluation monoculture. This dissertation does not compete as an RL method; it inherits the RL literature's problem framing (phase selection on a fixed decision interval) while rejecting its evaluation habits: I use measured topologies and demand, disjoint seeds with confidence intervals, and an architecture whose every decision is recorded with its stated rationale.

2.3 LLMs and SLMs for traffic signal control

[evidence: ANALYSIS.md \S 2 competitor map + \S 8 gaps]

The direct competitors apply language models to signal control, and the map is best read along three axes: where the model runs, how the evidence is powered, and what happens after a decision is made. LLMLight/LightGPT [19] is the closest prior work: GPT-4 rationales distilled into LoRA-tuned students down to 0.5B — but served on four A800-80GB GPUs, evaluated single-run on CityFlow with no seeds or confidence intervals anywhere in the paper. Traffic-R1 [31] trains a 3B model with GRPO and deserves real credit for a ten-intersection production deployment and for an ablation showing SFT degrading the base model's general benchmarks — yet its own future-work section concedes that edge deployment remains unrealised; this dissertation builds the artefact that paper only gestures at. DGLight [13] (a dissertation preprint, not peer-reviewed) tunes an 8B model on dual H100s and usefully calibrates the field: the *entire* LLM-versus-MaxPressure gap it and its peers report is roughly 1–2%, a margin that would not survive the seed variance I measure and that no competitor tests for. CoLLMLight [9] scales to 196 intersections on server GPUs and, tellingly, finds its fine-tuned 8B model beating a stock 671B model — distillation beating scale, which supports my central bet. CuraLight [10] is the one fine-tuning work on SUMO, but its 12B model needs roughly 44GB of VRAM against my 6GB budget. iLLM-TSC [26] inverts my safety topology — an RL agent proposes and a cloud GPT-4 vetoes — on a single synthetic four-way intersection, with a retry loop that falls back *silently* to the unvetted RL action; I return to this in §2.5, and I acknowledge its degraded-communications threat model as a scenario axis I do not test. LATS [20] distils an LLM teacher into MLP/GRU students — so nothing generative is ever deployed — and its pure-LLM baselines perform worst of all its methods, corroborating my finding that stock small models are weak deciders.

The pattern across all seven is consistent: every system serves from cloud or server GPUs; every headline table is single-run or, at best, $n \leq 10$ without statistical tests; none pairs control with any audit mechanism; and none evaluates on measured UK topologies. Those four absences are, jointly, the opening this dissertation occupies.

2.4 Small language models, distillation, and quantization

[evidence: ANALYSIS.md \S 4–\S 5]

I adopt the framing of the SLM survey [4] and the on-device review [6]: this system is *edge-only* (no cloud fallback), and I report the review's indicator template (time-to-first-token, VRAM, latency percentiles). Strikingly, neither survey contains a single traffic or control-loop case study — the application gap exists by omission. The nearest on-device comparator is Octopus [8], a 2B function-calling model at 1.1–1.7s on Android, bracketing my 0.3–1.5s decision latencies. The surveys also corroborate a reliability floor: Qwen2.5's own report [27] shows instruction-following (IFEval) collapsing from 84.1 at 72B to 27.9 at 0.5B, consistent with the 45–69% decision accuracy of my stock models. The Qwen3 report [28] complicates this — its 0.6B model posts respectable IFEval scores yet was my weakest stock decider; I resolved this in-house (the model parsed 93–100% strict JSON while choosing wrong phases), which sharpens rather than undermines the point: generic instruction-following benchmarks do not measure numeric decision quality, and the technical reports never test the latter.

On distillation, I must disclose a tension. Orca [24] argues that small models need rich explanation traces, whereas my distillation is label-only. Three defences: the task is a bounded, closed decision schema — closer to classification than open generation — and the knowledge-distillation survey [34] endorses labelling-plus-SFT for exactly this cell; Orca itself concedes small models excel in constrained settings; and empirically my student reaches near the teacher ceiling (97.7% against 98–99% teacher self-agreement). I concede what Orca implies: out-of-distribution generalisation and natively generated audit rationales are untested risks. The jump from 55% to 97.7% has precedent — distilled students outperforming their data-scale expectations [16, 29]. For the training recipe, LoRA [17] and QLoRA [12] justify rank 16, though their evidence stops above 125M parameters, and I flag the extrapolation. One precision matters: QLoRA trains adapters on a 4-bit base and serves unmerged; I train QLoRA-style, merge to bf16, then apply post-hoc int4 round-to-nearest for serving — two different quantisation steps, and QLoRA's authority does not transfer to the second. The quantisation literature in fact predicts trouble there: GPTQ [14] shows small models are the hard case for round-to-nearest, AWQ [22] finds the gap narrowing at 4-bit with grouping, and post-hoc quantisation after fine-tuning has been measured losing 9.4% relative even with GPTQ [5]. My contrary finding must therefore be scoped narrowly: bit-identical held-out decision accuracy under weight-only int4, on a near-converged narrow-schema checkpoint, measured with a discrete accuracy metric coarser than perplexity — a favourable open finding, not a literature-endorsed property. Finally, forgetting-scaling results [23] predict general-capability drift at my very low training loss, and show task accuracy and base-behaviour drift can diverge; I carry this as a declared limitation, doubly motivated by Traffic-R1's forgetting ablation. The gap this dissertation fills: no published work provides per-decision reliability data for a generative model at 0.6B, int4-quantised, inside a closed control loop — the surveys name size-versus-reliability as unresolved, and my measurements are new evidence on it.

2.5 Safe-by-construction shields

The canonical formalism is shielding [1]: a reactive system synthesised from an LTL safety specification and a finite MDP abstraction by solving a two-player safety game, placed either preemptively (the shield emits the safe-action set) or post-posed (the shield substitutes unsafe chosen actions). It carries proofs my system does not: correctness for all abstraction-consistent environments, minimal interference, and preserved learner convergence — and the authors' warning that the shield must remain active after learning, which mine does. Below this sit two rungs. Invalid-action masking [18] proves the masked update is a valid policy gradient and shows masking scales where penalties fail; agents trained masked remain largely valid unmasked. Neurosymbolic masking [7] learns the symbolic grounding when constraints are unknown — and its own future work points back towards temporal-logic specifications.

Positioned honestly, my shield is *not* a shield in the Alshiekh sense. It composes three weaker but sound pieces: preemptive action masking (the SLM selects from a fixed menu of pre-vetted conflict-free phase programs, so "no conflicting greens" holds by construction of the action set); a post-posed default-substitution filter (silent or invalid proposals are replaced by the deterministic MaxPressure choice); and an anti-starvation floor enforced by code rather than proved from a specification. What is missing is the LTL specification, the abstraction, and the game — hence no minimal-interference proof and no lookahead: my floor reacts *at* the starvation threshold where a synthesised shield acts before violation becomes unavoidable. The upgrade path is concrete and cheap at single-junction scale, and I name it as future work. This placement also explains why my topology beats iLLM-TSC's inversion [26]: they seat the least verifiable component (a prompted cloud LLM) in the safety-monitor position and fall back silently when it fails to parse, whereas I put the unverifiable component on the proposal side and keep the veto deterministic, local, and auditable — the only arrangement compatible with a 6GB edge premise. The gap: no prior LLM-based signal-control work states the formal status of its safety layer at all; this dissertation does, including what its shield is not.

2.6 Audit, transparency, and legal grounding

[evidence: 01-research/uk-traffic-law.md; ANALYSIS.md \S 7]

The dissertation's headline thread — that an edge controller should produce a provable account of its own decisions — sits in a three-way gap in the trust literature. On the attack side, Yu et al. [3] show that roughly 6% of colluding vehicles feeding falsified data to an undefended learning controller cut the colluders' waiting time by 92.5% while raising everyone else's by 62.7%; their most dangerous result is that a *calibrated* lie beats a maximal one, which directly motivates my stealth-liar scenario, and their paper explicitly names real-time anomaly detection as unbuilt future work. On the integrity side, the blockchain architecture of [2] rejects 100% of spoofed inputs at ~39ms — but never names a signature scheme, and its own future work concedes failure when the validators themselves collude; more fundamentally, input integrity is not decision accountability: it certifies what the controller *heard*, not what it *decided* or why. Nobody in this literature builds the defence.

My audit design therefore borrows its mechanism from outside traffic: Certificate Transparency [21] demonstrates that an append-only Merkle log with inclusion and consistency proofs makes tampering detectable without trusting the log operator — exactly the property a post-accident investigation needs, at a per-decision cost (a signature and a hash) that fits the edge budget where a validator network does not. The legal layer is deliberately modest. Dahl et al. [11] profile pervasive hallucination when language models answer legal questions, which rules out letting any model — let alone a 0.6B one — assert fault. My system instead emits a *cited evidence pack* against a UK-law fault-attribution knowledge base (§3.4): under it, authority-side liability is legally weak and arises essentially only from conflicting-green misfeasance, driver-fault doctrines are strong, and emergency-vehicle exemptions are conditional — so the artefact's job is to preserve the signed decision record that lets human lawyers apply those doctrines, never to output a verdict. The gap this dissertation fills is the strongest of the five it claims: it is the only system in this literature with a decision-level cryptographic audit trail, the only defence-side trust mechanism (a ledger in which truth-telling raises trust and detected lies are heavily penalised, tested against honest, blatant, and stealth adversaries), and consequently the only one that can produce a crash-to-UK-law evidence pack at all.

Chapter 3

System design

The system's design premise is that a learned controller at a junction must be *optional for safety and mandatory for accountability*: the deterministic control path must be complete without the model, while every actuation — whoever authored it — must be provable afterwards. This chapter presents the architecture that realises the premise, and defines the two structural quantities (override rate, authored share) whose measurement becomes the dissertation's central finding.

3.1 Architecture overview

Each junction runs an identical control loop on a fixed decision interval ($\approx$10 s simulated). Per tick: the junction's local state (per-phase queue and waiting measurements from its own sensors) is rendered into a prompt; the on-device SLM, served by Foundry Local, proposes a phase; a deterministic parser extracts the proposal; the safety shield masks, validates, and possibly substitutes it; the served phase actuates the lights; and a signed audit record binding state, proposal, served action, and authorship is appended to the junction's hash chain. The MaxPressure policy is computed every tick regardless of the SLM — it is simultaneously the shield's reference, its fallback, and the strong baseline — so the marginal cost of the learned component is one model call, and its marginal risk is bounded by construction.

Three design consequences follow. First, graceful degradation is the default: if the model times out, emits garbage, or the serving runtime dies (all observed in practice, §6.8), the junction continues under MaxPressure with the failure recorded. Second, the architecture inverts iLLM-TSC's topology [26] — there, an RL agent proposes and a cloud LLM vetoes; here, the LLM proposes and a local $O(\text{phases})$ verifier vetoes — which is the only orientation deployable at the edge, since the veto path must be cheap, local, and always available. Third, because the deterministic path is complete, every claim about the SLM's value can be measured as a *difference* against the same loop with the proposer disabled.

3.2 The SLM decision agent

The agent contract is deliberately minimal: the model receives a system instruction and a rendered state, and must return exactly `{"phase": N}`. Temperature is zero; the serving endpoint is OpenAI-compatible; parsing falls through four tiers (strict JSON, braced, keyed, bare integer), with the tier recorded per decision so format discipline is measurable, not assumed.

Four prompt formats act as graded capability probes, each revealing a different competence. **Myopic** exposes only per-phase halting counts — the least privileged view, answerable by a counting heuristic. **Sota** adds delay-aware context (accumulated waiting), the state a delay-optimal policy actually needs. **Prediction** adds short-horizon arrival forecasts, probing whether the model can use anticipatory information. **Coordination** adds neighbour notes, probing multi-junction awareness. The formats are held identical between offline evaluation and closed-loop control — the same rendering code serves both — so offline accuracy numbers transfer interpretably to the loop.

3.3 The safety shield

The shield composes three mechanisms, named here with the vocabulary of the formal literature (§2.5) so that what it is *not* is explicit:

1. **Preemptive action masking** [18]: the model chooses from a fixed menu of conflict-free phases; no proposal can create a conflicting green, because conflicting actions are not in the action space.
2. **Post-posed default substitution**: an invalid, unparseable, or absent proposal is replaced by the

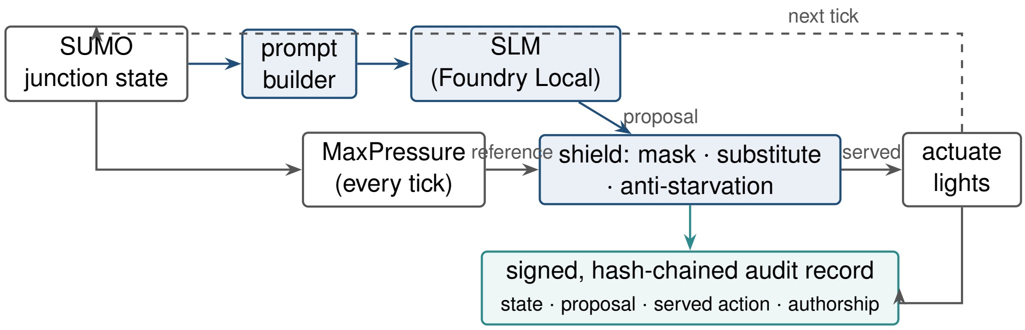

Figure 3.1: The junction control loop. The deterministic path (grey: state → MaxPressure → shield → actuation) is complete without the SLM; the learned proposer (blue) attaches to it, and every served decision — whoever authored it — lands in the signed audit chain (teal).

MaxPressure phase. The substitution is recorded as such — it is a served decision the SLM did not author.

3. **Anti-starvation runtime monitor**: a minimum-service floor guarantees every approach is served within a bounded window, overriding both the SLM and MaxPressure if necessary.

This is a sound heuristic composite, not a correct-by-construction shield in the Alshiekh sense [1]: there is no temporal-logic safety specification, no game-theoretic synthesis, and no minimal-interference proof. I state this as a limitation with an upgrade path (§7.4); the composite nonetheless guarantees the two properties the evaluation depends on — no conflicting greens, no starved approach — for every arm including the baselines.

Two measured quantities are defined on this structure, and the precision matters because Chapter 6 turns on them. **Override rate** is the fraction of decision ticks where the SLM's valid proposal diverged from the MaxPressure reference and the reference was served instead. **Authored share** is the fraction of all served actuations whose phase originated with the SLM (whether or not it agreed with MaxPressure). Neither is a veto count of unsafe actions — masking makes unsafe proposals impossible — they are *interference* measures: how much of the deployed behaviour is attributable to the learned component rather than its classical scaffold.

3.4 Audit trail and crash reconstruction

The audit layer is designed backwards from the question an investigator asks after a collision: *what did the controller know, what did it decide, who authored the decision, and can the record be trusted?*

Every decision tick appends a record containing the state digest, prompt hash, model identity and version, raw proposal, parse tier, served action, authorship (SLM / shield-substitution / anti-starvation), and latency. Records are Ed25519-signed and hash-chained per junction; chain verification and signature verification run end-to-end after every experiment, and the threat model includes an imposter re-signing attack (caught in all evaluated scenarios, §6.6). The chain head can be anchored to external witnesses in a Certificate-Transparency-style quorum [21]; a permissioned-ledger anchor is implemented as one optional witness, deliberately positioned as an anchoring option rather than a contribution.

On top of verified records sits the reconstruction layer: a UK-traffic-law knowledge base (statutes, case law, the conditional emergency-vehicle exemption) against which the verified facts of a collision scenario are matched deterministically. The output is a *cited evidence pack* — the governing rules, the verified facts that trigger them, and the fault anchor each rule implies — never a verdict. This design choice is a direct response to the legal-hallucination literature [11]: the system's legal knowledge is a curated, citable database, and the adjudicative act is explicitly left to humans. The dissertation's own scenarios use synthetic actors on real topology, so no real data subject exists; the residual legal exposure of a real deployment is analysed in §7.3.

The audit design is also why authorship (§3.3) is the pivotal metric: an evidence pack over a deployment whose decisions are mostly shield substitutions attributes the behaviour to a textbook algorithm, not to the certified learned component. For the audit to be *about the model*, the model must author the decisions — which is precisely what fine-tuning turns out to buy (§6.5).

3.5 Trust-coefficient mechanism

Coordination requires acting on neighbour reports, which is exactly what a compromised neighbour can abuse. The trust mechanism prices that risk with three commitments. First, **local sensing is the ultimate truth**: a junction's own sensors are never doubted, and every neighbour claim is eventually scored against what local sensing later verified. Second, **asymmetric update**: verified truths earn trust slowly (prior 0.5, ten consecutive truths to reach 0.9) while a detected lie from a high-trust source collapses it in one step (lie factor 0.25), because in infrastructure the cost of trusting a liar exceeds the cost of doubting an honest peer. Third, **discount-only wiring**: trust can strip a caught liar's power to corroborate a preemption, but can never *add* an action on zero local evidence — so the mechanism cannot regress safety, and a real ambulance is always preempted via the junction's own sensing. A truthful report that local sensing verifies late is vindicated retroactively, so honesty is never permanently penalised for latency. The adversarial evaluation (honest, blatant, stealth misreporting injected into closed-loop SUMO) and its honest partial results are reported in §6.7.

3.6 The fine-tuning axis

The final design axis asks what changes when the proposer is fine-tuned rather than stock. The pipeline is distillation: a frontier teacher labels junction states sampled from training-reserved seeds; labels are cross-audited against a second frontier model (98–99% agreement); a deterministic sanity verifier screens states, not answers, so the teacher is never second-guessed by a weaker rule; and the four prompt formats are rendered from the same labelled pool into one mixed training set. Training is QLoRA [12] (r=16, NF4 double quantisation, completion-only loss, family-correct chat templates); the merged bf16 model is compiled to weight-only int4 ONNX and served through the same Foundry Local runtime as every stock model.

Three design rules protect the comparisons. The holdout is hash-frozen before any training (§5.3); closed-loop evaluation seeds are disjoint from training seeds; and the control arm is the same base model compiled through the identical int4 pipeline, so the only manipulated variable is the weights. Ablation students — leave-one-topology-out and four format specialists — are trained under the same recipe to test memorisation and specialisation hypotheses directly rather than by argument.

Chapter 4

Implementation

This chapter records the engineering that made the design measurable: the simulated London networks, the on-device serving stack and its documented failure modes, the distillation and compile chain, and the experiment harness. I keep it tight and evidence-focused; the reproducibility pins (versions, seeds, templates) are tabulated in Appendix A.

4.1 Simulation environment and topologies

All control runs in SUMO with TraCI. Six topologies span the regular-synthetic to irregular-real axis that Chapter 6 shows is decisive: two synthetic grids (3×3, 4×4); the Euston A501 corridor and its peak-hour variant, whose demand uses the measured DfT hourly diurnal profile and vehicle mix for the A501; a Bloomsbury grid extracted from OpenStreetMap; and the Old Street complex junction (nine signals including a single five-arm, seventeen-movement junction, demand-calibrated after default random demand gridlocked it). Real-map demand magnitudes are measured; their vehicle-by-vehicle realisations are synthetic draws per seed — a scoping honestly carried through every claim (§7.3).

4.2 On-device serving with Foundry Local

Serving is Microsoft Foundry Local (pinned 0.8.119) on an RTX 2060 with 6 GB VRAM — consumer hardware of roughly signal-cabinet class, and a hard project constraint rather than a convenience: the model ladder is capped at ~4B (7B-class models OOM the runtime, recorded), which is what makes the scale-saturation result (§6.3) deployment-relevant rather than academic.

Custom models serve through two execution providers. The CPU path works out of the box; the WebGPU path requires editing the compiled model's `genai_config.json` provider options — and delivers roughly 2× lower latency with *bit-identical* holdout decisions, which licensed running every sweep GPU-served. Windows ML CUDA autoregistration requires a newer Windows build than the test machine's, making WebGPU the effective GPU path; I document this because none of it is in the product documentation, and the recipe is part of the contribution.

Sustained unattended serving surfaced four reliability findings — catalogue-artifact token corruption after long uptime, two unprompted service self-restarts mid-evaluation, and undocumented registration requirements for compiled models — each mitigated in harness (§4.5) and reported as results (§6.8).

4.3 Distillation dataset and training

From four training-reserved seeds I sampled 5,952 unique junction states and labelled them with a frontier teacher; a second frontier model re-labelled audit samples at 98.0–99.0% agreement. Rendering the four prompt formats over the labelled pool yields 59,533 training examples and a hash-frozen holdout of 5,580 (`HOLDOUT_MOD=12`). Rendering is family-aware — ChatML with an empty think block and no-think marker for Qwen3; Qwen2.5's ChatML without the marker; Phi's `<|system|>...<|end|>` convention — with loss computed on the completion only, targeting exactly `{"phase": N}`.

Training is QLoRA r=16, NF4 double quantisation, fp16 compute. The 0.6B student trained on the serving machine itself (13.5 h on the RTX 2060 — the full loop from teacher to served controller fits on one consumer device); the 3.8B student and its five ablation variants trained on a single UCL RTX 4090 (2.3 h each, 8.9M trainable parameters, 0.23%).

4.4 Compile-and-serve chain

The path from merged bf16 weights to a served endpoint is undocumented for custom models, and assembling it produced pinned lessons now recorded in the project's feasibility log: the onnxruntime-genai model builder must run against a specific transformers pairing (newer tokenizer configs emit a `tokenizer_class` Foundry cannot load — patched to the catalogue value); Phi checkpoints need their remote-code mappings stripped; and a compiled model is not servable until an inference manifest (`inference_model.json` — name and prompt template) is *synthesised*, because the builder does not emit one. The eventual chain — QLoRA merge → int4 RTN compile → config patch → manifest synthesis → cache registration → WebGPU provider swap — is, to my knowledge, the first documented end-to-end recipe for serving a custom fine-tuned model through Foundry Local.

4.5 Experiment harness

Two harnesses produced every number in this dissertation. The **factorial runner** executes model × format × topology × seed cells with per-cell raw JSON artifacts, a pre-flight serving probe per cell (no silent parity: an unservable model records a skip, never a score), resumability (existing cells are skipped, so crashes and restarts lose at most one cell), and multi-pass retry for deferred cells. The **offline evaluator** scores holdout rows against the served endpoint with fail-loud error accounting — API errors are counted, never scored as answers — and, after the observed service self-restarts, recovery logic that rediscovers the endpoint, reloads the model, and retries failed rows with statistics intact. The harness survived a six-week unattended campaign including two service restarts, one machine reboot, and one mid-write power loss (one corrupted cell, detected by JSON validation and re-run); I regard the fail-loud discipline itself as a methodological contribution of the work.

Chapter 5

Experimental methodology

This chapter defines what I measured, how each comparison was controlled, and the statistical decision rules I committed to before running the experiments. The methodological stance throughout is fail-loud: harness errors abort or are counted, never silently scored, and null results are reported as findings.

5.1 Evaluation axes and metrics

I evaluate along four axes, each with its own primary metric.

Closed-loop control quality. The primary metric is mean network delay: the mean, over all vehicles loaded into the simulation, of the excess travel time relative to free-flow, measured over a fixed 1,200-step horizon. Every controller cell is paired with a fixed-time baseline run on the *same topology and seed*, so the reported quantity is the paired relative delay change. Secondary metrics — completed-vehicle throughput, teleport count, and undeparted backlog — guard against the classic failure mode of delay metrics, where a controller "improves" delay by starving demand.

Offline decision quality. Accuracy against frozen frontier-teacher labels on a held-out set of junction states, at int4 precision exactly as served. I also report strict-JSON rate (the fraction of responses matching `{"phase": N}` exactly) and parse rate (any recoverable phase), because a controller that cannot keep its output contract imposes a different kind of cost than one that chooses a wrong phase.

Structural authorship. Two quantities defined in §3.3: override rate — the fraction of served SLM decisions where the proposal diverged from MaxPressure and the shield's choice was served instead — and authored share, the fraction of all actuations whose served phase originated with the SLM. I treat these as first-class results rather than diagnostics, because the audit argument of this dissertation (§3.4) turns on them.

Serving viability. Decision latency (p50/p99) against the 10 s decision interval, and a log of serving-reliability events with their mitigations.

5.2 Statistical protocol

Every closed-loop cell is replicated over n=30 random seeds per topology. The test for a delay effect is a paired t-test on per-seed (controller, baseline) delay pairs; I additionally pool across topologies (paired, n=90) where the hypothesis is topology-independent. p-values are computed with the project's pure-mathematics implementation of the regularised incomplete beta function — the same code path used by the two-way ANOVA of the explanatory study — so no statistics library version can silently change a reported number. For the factorial explanatory analysis I report effect sizes (η^2) alongside F statistics, and for headline comparisons I state the direction, magnitude, and p together rather than significance alone.

Two commitments were made in advance. First, the honest-null policy: a non-significant result at n=30 is reported with its magnitude and uncertainty, not suppressed or re-run until significant. Second, no mid-experiment stopping: cell counts were fixed by the factorial design before launch, and the resumable runner (§4.5) re-runs only cells that produced no measurement (crash, probe failure), never cells whose answer was unwelcome.

5.3 Seed-disjointness and holdout freezing

Two leakage paths exist between training and evaluation, and each is closed by construction.

State-level leakage. The distillation dataset is built from junction states sampled on four reserved seeds

Table 5.1: Experiment inventory. Every family retains per-cell raw JSON.

Family	Design	Primary output
Base-model factorial	8 catalogue SLMs × 4 formats × 6 topologies × up to 32 seeds	closed-loop delay vs baselines
Explanatory (why) study	2×2 factorial ANOVA over model/config on 4 topologies	F, η^2, interaction
Offline distillation	7 phi rows + 4 qwen rows × 4 formats, frozen holdout	accuracy, JSON rate
Fine-tune closed-loop	ft vs stock × 3 topologies × 30 disjoint seeds (both scales)	paired delay, authorship
Decision battery	cloud frontier vs local SLM on identical states	agreement, latency
Accident / crash-law suite	scripted collision scenarios → audit reconstruction	evidence packs, chain verification
Trust battery	honest / blatant / stealth misreporting	detection, delay impact
Emergency / EV suite	EV arrival scenarios per topology	clearance metrics

($\{3, 7, 11, 29\}$). The offline holdout is split from the same state pool by a deterministic hash of the state content (`HOLDOUT_MOD=12`, giving 5,580 of 65,113 rendered examples), and was frozen — file hash recorded — before the first training run. No training example, under any format rendering, shares a state with a holdout example.

Trajectory-level leakage. Closed-loop evaluation uses seeds $\{1..30\} \setminus \{3, 7, 11, 29\} \cup \{31, 32, 33, 34\}$ — thirty seeds fully disjoint from the training seeds. This matters because a controller fine-tuned on states from seed s has effectively seen a compressed replay of seed s's demand realisation; evaluating on it would overstate generalisation. All closed-loop numbers in Chapter 6 are disjoint-seed numbers.

5.4 Baselines and controls

Fixed-time is the floor every deployed junction already achieves: a cyclic plan with equal splits, the same phase menu as every other arm. **MaxPressure** is the theoretically-grounded adaptive baseline [30], and doubles as the shield's fallback policy.

The control for every fine-tuning comparison is the *same base model compiled through the identical int4 pipeline* (denoted stock, or ft0), not the vendor catalogue artifact. This choice was forced by measurement: the catalogue Qwen3-0.6B GPU artifact was observed emitting corrupted tokens after long service uptime (a documented reliability finding, §6.8), and a catalogue artifact differs from my compiled one in quantisation recipe even when healthy. Holding the compilation pipeline fixed isolates the single manipulated variable — the fine-tuned weights.

Two further controls target specific hypotheses. A **leave-one-topology-out** (LOTO) student, trained with all Old Street states removed, tests whether distillation gains are topology memorisation. Four **format-specialist** students, each trained on one prompt format alone, test whether a deployed junction would prefer a per-format model over a generalist; the answer (§6.3) was no at both model scales.

5.5 Experiment inventory

Table 5.1 maps every experiment family reported in Chapter 6 to its design and artifact trail. All raw cells, per-seed JSON records, and aggregation scripts are retained in the project repository; every number in this dissertation is recomputable from them.

The full campaign spanned three machines (the RTX 2060 serving/measurement node, a UCL RTX 4090 for 3.8B training, and an RTX 4060 reserve) and roughly six weeks of wall-clock compute; the factorial policy — capture every cell, gate nothing on interim results — follows the project's standing directive that the dissertation should be able to answer questions I had not yet thought to ask when the sweeps were launched.

Chapter 6

Results

I present results in the order the evidence was built: the baseline landscape that reframed the project (§6.1), the base-model factorial (§6.2), the offline distillation tables (§6.3), the closed-loop verdict (§6.4), the authorship finding this dissertation headlines (§6.5), the audit and trust results (§6.6, §6.7), and the serving-reliability evidence (§6.8).

6.1 Baseline landscape: the MaxPressure reversal

Adding a naive fixed-time floor to every comparison produced the single most consequential rigour result of the project. MaxPressure, throughput-optimal in theory [30] and dominant in the simulation literature, is *worse than naive fixed-time on every measured London topology in this study* — Euston +15%, Old Street +5% (one seed +37%), Bloomsbury +3% — while dominating exactly where the literature tests it, on regular synthetic grids (−27% on grid4×4, −19% on grid3×3).

I scope this claim deliberately: six independent studies in my corpus report MaxPressure winning on grid networks, and nothing here contradicts them. What the reversal shows is that its guarantees do not transfer to irregular real-world geometry, and the mechanism is visible in the theory itself: the throughput-optimality proof assumes infinite queue capacity, and finite-capacity extensions prove work-conservation losses [15], while stability results hold under heavy load [35] — London arterials at measured demand sit in neither regime. The consequence for evaluation practice is immediate: any SLM-vs-MaxPressure win on real topologies is partly MaxPressure's own weakness, so every claim in this dissertation is stated against the fixed-time floor, with MaxPressure reported alongside.

6.2 Base-model factorial

The full factorial (seven servable catalogue models — the 6 GB card caps the ladder at ~4B, with 7B-class OOM failures recorded — × four prompt formats × five topologies) established three facts before any fine-tuning.

First, shielded stock SLM control beats the fixed-time floor on every topology (Euston −16%, Bloomsbury −2%, Old Street −14% (seed-noisy), grid3×3 −21%, grid4×4 −28%), and beats MaxPressure specifically where MaxPressure breaks down — the real topologies. Second, results must be conditioned on topology: pooling across maps is a Simpson's-paradox trap, since corridor wins and early grid blow-ups average into a meaningless mean. Third, per-decision reasoning quality does not predict closed-loop value: on 80 identical junction states, cloud frontier models agree with the delay-optimal heuristic at 92–96% against 51–72% for local SLMs, yet a near-chance local model still won its closed-loop cells. The controller's value is systemic — the SLM, shield, and event gating together — not raw single-junction reasoning. This decision battery is open-loop by construction (the frontier teacher cannot be called from inside the simulation loop) and is stated as such.

The explanatory two-way ANOVA over the factorial confirms the structure: model and configuration interact strongly (interaction F=32, p<0.001, η^2=0.86), with the effect driven by outlier cells rather than a smooth gradient — a further argument against pooled claims.

The powered stock-model replication (n=30 per cell, four catalogue models × two formats, both baselines) sharpens the picture per topology. On **Old Street**, stock SLM control achieves the campaign's strongest powered wins against the fixed-time floor — Qwen2.5-0.5B sota −11.8% (p<0.001), Qwen3-0.6B sota −9.6% (p<0.001), Phi-4-mini myopic −8.6% (p<0.001) — with one arm also significantly beating MaxPressure (Qwen2.5-0.5B sota, −4.9%, p=.014). On **Euston peak-hour** under the measured DfT demand profile, every arm is statistically indistinguishable from MaxPressure and none beats the floor:

Table 6.1: Holdout accuracy (%) against frozen frontier-teacher labels, int4-as-served. All fine-tuned rows and the stock Phi row emit 100% strict JSON; stock Qwen3-0.6B reaches 93.4% on sota.

	myopic	sota	prediction	coordination
Qwen3-0.6B stock	45.6	55.1	63.3	69.3
Qwen3-0.6B generalist ft	**87.9**	**97.7**	**97.6**	**99.0**
Phi-4-mini stock	87.5	59.0	86.0	87.0
Phi-4-mini generalist ft	**88.7**	**97.1**	**97.4**	**99.3**
Phi-4-mini LOTO	88.3	96.8	97.4	99.1
Phi-4-mini sota-specialist	87.9	97.8	89.5	93.3
Phi-4-mini myopic-specialist	87.0	87.9	88.8	91.8
Phi-4-mini prediction-specialist	87.9	91.3	97.3	98.4
Phi-4-mini coordination-specialist	87.4	89.1	94.7	99.3

the corridor null is a powered result, not an underpowered one.

The replication also yielded a methodological finding I had not planned to measure. On the 26 seeds shared between runs, the *catalogue* Qwen3-0.6B artifact and my identically-prompted, identically-shielded pipeline compile of the same weights differ by nine points of mean relative delay (+4.1% vs −5.2% against the floor) on this high-variance topology — quantisation and packaging provenance alone shift the headline sign. This is precisely why every fine-tuning comparison in this dissertation uses the identical-pipeline compile as its control (§5.4), and it is a caution for the field: "the same model" is not the same controller across serving artifacts.

6.3 Offline distillation: saturation at 0.6B

Table 6.1 gives the complete offline campaign: both student scales, stock versus fine-tuned through the identical int4 pipeline, plus the LOTO and format-specialist controls, evaluated on the frozen 5,580-state holdout as served (WebGPU, int4).

Five findings follow.

1. **Scale saturation.** The 0.6B and 3.8B generalists are statistically indistinguishable on every format, both sitting at the teacher-agreement ceiling (98–99% teacher self-consistency). Distillation saturates this decision task at 0.6B; six times the parameters buy nothing per-decision.
2. **Where the gain lands depends on the base.** The 0.6B is lifted +32 to +42 points everywhere; Phi-4-mini, already 86–88% on three formats stock, gains materially only on the delay-aware sota format (+38.1 points). For a capable base, fine-tuning buys decision quality, not format discipline — stock Phi is already 100% format-compliant.
3. **Generalists only.** No format specialist beats the generalist on its own format by more than 0.65 points; all pay 4–8 points off-format; and the myopic specialist loses *even on-format* (87.0 vs 88.7), consistent with mixed-format training regularising against the noisy privileged-teacher labels that pure-myopic training overfits. The result replicates across both model scales.
4. **No topology memorisation.** The LOTO student, trained with all Old Street states removed, matches the generalist on every format.
5. **Quantised serving is exact for this task.** CPU and WebGPU int4 artifacts agree bit-identically (97.42% both on the sota holdout); I scope "lossless" strictly as bit-identical holdout accuracy under weight-only int4 RTN.

The myopic ceiling (~88%) is intrinsic: the teacher was privileged with waiting-time state the myopic prompt hides, an Orca-style label-only distillation cost [24] disclosed as a design property, not noise.

Table 6.2: Closed-loop mean network delay vs fixed-time (n=30 disjoint seeds per cell; paired t). Negative favours the controller.

Topology	Arm	Δdelay	p	override	authored
Euston peak-hour	ft	−3.9%	.16	0.12	0.70
	stock	−6.2%	.12	0.47	0.68
Bloomsbury grid	ft	+0.7%	.37	0.18	0.73
	stock	+2.6%	1.5×10^{-3}	0.46	0.65
Old Street	ft	−3.1%	.20	0.11	0.90
	stock	−1.1%	.35	0.60	0.54
Pooled (n=90)	ft	−2.1%	.14		
	stock	−1.6%	.39		
Euston	phi ft	−3.8%	.15	0.13	0.70
Bloomsbury	phi ft	+0.6%	.42	0.17	0.73
Old Street	phi ft	−1.0%	.56	0.12	0.88
Pooled (n=90)	phi ft	−1.4%	.20		

6.4 Closed-loop verdict: delay decouples from accuracy

Table 6.2 reports the Qwen3-0.6B closed-loop sweep: fine-tuned (ft) versus identically-compiled stock, three topologies × 30 training-disjoint seeds, paired per-seed against fixed-time.

The honest headline: at n=30 per topology, neither arm significantly beats the fixed-time floor anywhere, and a +42-point offline accuracy gain produces no significant delay change — the **delay-decoupling paradox** anticipated by the decision battery. The one significant effect of fine-tuning is protective: stock control significantly *harms* the congested Bloomsbury grid (+2.6%, p=1.5×10^{-3}), the fine-tuned controller is neutral there, and the head-to-head paired difference favours fine-tuning by −1.8% (p=.019). I return to the mechanism — the shield bounds worst-case behaviour for both arms, and external calibration puts the entire LLM-vs-MaxPressure headroom at 1–2% [13] — in §7.2. The Phi-4-mini confirmation arm (final rows of Table 6.2) extends the saturation finding end-to-end: the 3.8B fine-tuned student's closed-loop profile is arm-for-arm indistinguishable from the 0.6B's — Euston −3.8% vs −3.9%, Bloomsbury +0.6% vs +0.7%, the same override band (0.12–0.17) and authored share (0.70–0.88) — at p99 latency still ≤1.1 s on the 6 GB device. The dissertation's scale-threshold question is thereby answered in both evaluation regimes: 0.6B suffices.

6.5 Authorship conversion

The structural columns of Table 6.2 carry the dissertation's central finding. Fine-tuning drops the override rate from 0.46–0.60 to 0.11–0.18 on every topology and raises the authored share of served decisions to 0.70–0.90. The stock deployment's delay numbers are substantially the *shield's* numbers: most of what the stock model proposes diverges from MaxPressure and is not served. The fine-tuned model's numbers are its own.

In the interference vocabulary of the shielding literature [1, 18], fine-tuning internalises the shield's envelope: the model learns to propose actions that survive. I argue this, not delay, is the deployable product of distillation for an audited edge controller. An accident audit over a stock deployment attributes the served decisions to the fallback policy; the same audit over the distilled deployment attributes them to the certified, versioned model the operator can retrain and re-certify. Causal accountability is what moved, and it moved by a factor of four.

Figure 6.1: Authorship conversion (n=30 seeds per bar). Fine-tuning cuts the override rate roughly four-fold on every topology (red) and raises the SLM's authored share of served decisions (teal), most sharply on Old Street (0.54 → 0.90).

6.6 Audit and crash reconstruction

The audit layer was evaluated on seven pre-defined collision scenarios spanning the legally distinct cases (red-light running, EV exemption, maintenance vehicle, amber gambling, conflicting green, dark signal, spoofed EV). All seven prove end-to-end: hash-chain verification passes, signature verification passes, and an imposter re-signing attack is caught in every scenario. The governing UK rule is *inferred deterministically from the verified facts* — e.g. the spoofed-EV scenario correctly denies the exemption and anchors fault on the driver, while the conflicting-green scenario is the sole case attributing (weak, misfeasance-only) authority fault — and the system emits a cited evidence pack for a human adjudicator, never a binding verdict, by design [11]. [TODO: one-paragraph worked example of a single evidence pack, from audit record to cited rule, as a figure or listing.]

6.7 Trust-coefficient battery

The trust mechanism passes its full property battery (9/9): a persistently honest junction climbs to 0.929 trust; a persistent liar collapses to 0.011 and loses the power to corroborate; the asymmetry is deliberate (ten verified truths to climb from 0.5 to 0.9, one lie from high trust falls 0.65); and a truthful-but-late report is vindicated retroactively by local sensing, so honesty is never permanently penalised for latency. Trust is discount-only by construction — it can strip a liar's corroboration power but can never add a preemption on zero evidence, and local sensing is never itself doubted.

In closed-loop traffic (honest / blatant / stealth misreporting injected into SUMO), the mechanism detects and contains blatant lying on grid networks, while a stealth attacker calibrated to stay within plausible bounds evades detection, and the mechanism is inert on the corridor (where coordination traffic is minimal). I report the stealth evasion as an open limitation rather than defending against it post-hoc. [TODO: exact detection/delay numbers from experiment_trust_traffic results.]

6.8 Serving reliability at the edge

Sustained on-device serving produced four documented reliability findings across the campaign: (i) the catalogue Qwen3-0.6B GPU artifact emitting corrupted tokens after long service uptime; (ii, iii) two unprompted Foundry Local service self-restarts mid-evaluation, killing runs at 5,500/5,580 and 4,300/5,580 rows; (iv) registration of builder-compiled models requiring undocumented synthesis of the inference manifest. Each was mitigated in harness — endpoint rediscovery, model reload with retry-of-failed-rows and statistics intact, fail-loud parse accounting that aborts rather than scores error text — and the mitigations are themselves contributions: they are what turned a research prototype into a harness that survived a six-week unattended campaign. Decision latency held at 0.4–1.1 s p50 GPU-served across all models — a 9× margin inside the 10 s decision interval — and p99 stayed under 2 s in every sweep.

Chapter 7

Discussion

7.1 Answering the research questions

RQ1 (edge competence): yes, with honest scope. Shielded on-device SLM control beats the fixed-time floor on every topology tested and beats MaxPressure precisely where MaxPressure's assumptions fail — real irregular geometry. But at powered replication (n=30 disjoint seeds) the per-topology delay effects are not individually significant, and the corridor null persists across every controller variant. The defensible claim is competence within the deployable envelope, not superiority.

RQ2 (what distillation buys): authorship, and it saturates at 0.6B. Offline, distillation is dramatic (55–69% → 97–99%) and scale-flat — a 3.8B student adds nothing over 0.6B, and no format specialist beats the generalist even on its own format. Closed-loop, the accuracy does not become delay; it becomes *ownership*: override falls four-fold, authored share reaches 0.90, and the one significant stock harm (congested grid, +2.6%, p=1.5×10^{-3}) is repaired to neutral. For a shielded, audited controller I argue this is the correct success criterion, and §7.2 explains why delay was never the sensitive variable.

RQ3 (provable audit): yes, as designed. Seven scenario reconstructions prove end-to-end with tampering caught; the system emits cited evidence packs, and the legally weak branches (authority fault) are weak *in the law*, faithfully represented rather than overstated.

RQ4 (trust): partial, honestly. The mechanism's properties all hold, blatant misreporting is contained, but a calibrated stealth attacker evades detection and the mechanism is inert where coordination traffic is minimal. A trust ledger is a necessary layer, not a sufficient defence.

7.2 Why accuracy decouples from delay

Three mechanisms jointly explain the paradox, and all three are visible in the data rather than conjectured. First, *the shield compresses the outcome distribution*: masking, substitution, and anti-starvation bound the worst case for every arm, so the behavioural difference between a 59% and a 97% proposer is far smaller served than raw. Second, *the headroom was small*: external calibration from a 13× larger tuned model puts the entire LLM-vs-MaxPressure delay gap at 1–2% [13], and my own decision battery showed a near-chance proposer winning closed-loop cells — per-decision agreement with a delay-optimal heuristic is simply not the trajectory-level objective. Third, *where the stock model could do damage, fine-tuning shows up*: the single significant closed-loop effect is the repair of stock harm under grid congestion, exactly where bad proposals compound fastest. The practical reading for deployment: fine-tune to own the decisions and to remove harm tails, and do not expect — or promise — delay headlines inside a safety shield.

7.3 Threats to validity

I declare four threats with their defences, and none is cosmetic.

Label-only distillation (construct). I distil teacher labels, not reasoning traces; Orca-style trace distillation [24] might transfer more competence. The myopic format makes the cost measurable: its ~88% ceiling is the privileged-teacher gap, disclosed as a design property.

Retention (internal). Aggressive SFT can erode general capability with scale-dependent forgetting [23]. I measured it: a seeded 200-item MMLU sample served through the same endpoints shows the 3.8B student dropping from 57.5% (stock) to 53.0% (fine-tuned) — a decline consistent with mild forgetting but not significant at this sample size (z=0.91, p=.37) — while retaining its answer-format discipline (94% strict JSON). At 0.6B the probe is uninformative by floor effect: the stock model already scores at chance

(26.0%) on this format, so there was no measurable general capability to lose. The honest conclusion is that no significant retention damage was detected, with the caveat that a 200-item probe bounds only large effects.

Quantisation scope (construct). "Lossless" int4 is claimed strictly as bit-identical holdout accuracy under weight-only RTN — not as a general equivalence claim, which the quantisation literature would not support [5, 14].

Simulation and demand realism (external). Everything is SUMO; real-map demand magnitudes are measured (DfT hourly profile on Euston) but their vehicle-level realisations are synthetic draws, and comparators exclude trained deep-RL systems. The MaxPressure reversal is accordingly scope-bound to measured London topologies — six grid studies showing the opposite are cited, and my own grids reproduce their direction. Legal scope: the audit scenarios use synthetic actors on real topology, so no real data subject exists; a real deployment would face the adjudication and disclosure exposures analysed in the project's legal grounding, which curation itself is an adjudicative act (*Bates v Post Office (No 6)*) and qualified privilege is defeasible.

7.4 Limitations and future work

Four directions follow directly from the findings. **Formal shield synthesis**: replace the heuristic composite with a synthesized shield from a temporal-logic safety specification [1] — the per-junction product game is small enough for exact synthesis, and the authorship metrics defined here would then carry minimal-interference guarantees. **Closed-loop objectives**: GRPO-style tuning against trajectory reward [13] attacks the decoupling directly; label distillation is the wrong objective if delay is the goal. **Stealth-robust trust**: detection calibrated to distributional implausibility rather than single-report bounds. **Hardware-in-the-loop**: the serving-reliability findings (service self-restarts, artifact corruption) are exactly the failure class a cabinet deployment must survive, and they argue for a watchdog-supervised serving architecture before any field pilot.

Chapter 8

Conclusion

This dissertation asked whether an on-device small language model, served by Microsoft Foundry Local within a 6 GB envelope, can control real London junctions competently while producing an accident audit worth trusting — and at what scale the learned component earns its place.

The answer to feasibility is yes, with the scope stated plainly. A shielded SLM beats the fixed-time floor on every topology tested, and beats MaxPressure exactly where its optimality assumptions collapse — on measured, irregular London geometry, where I show the celebrated baseline is worse than naive fixed-time. The full control loop, from frontier-teacher distillation through QLoRA training to int4 serving, fits on one consumer device; the 0.6B student trained on the same GPU that serves it.

The answer to scale is sharper than expected: distillation saturates at 0.6B. A student six times larger lands on statistically identical decision accuracy, no format specialist beats a generalist even on its own format, and a student trained without one topology loses nothing on it. Within this deployable envelope, capability is not the bottleneck — the distillation ceiling is.

The central finding reframes what fine-tuning is *for* in a shielded controller. Closed-loop, at thirty disjoint seeds per topology, the 97%-accurate student does not significantly beat the floor — and neither does anything else inside the shield's envelope, whose compression of outcomes, together with the small delay headroom the wider literature itself reports, explains the decoupling. What fine-tuning demonstrably moves is authorship: override falls from 0.46–0.60 to 0.11–0.18, the model's authored share of served actuations reaches 0.90, and the stock model's one significant harm disappears. Because the audit layer proves who decided what — seven collision scenarios reconstructed end-to-end, tampering caught, governing UK law inferred from verified facts into cited evidence packs — authorship is precisely what makes the audit about the certified model rather than its fallback. Fine-tuning buys accountability, not delay.

The dissertation's credibility rests as much on its nulls as its wins: the corridor null at n=30, the stealth attacker that evades the trust ledger, the privileged-teacher ceiling on the myopic format, and the four serving-reliability failures are reported with mechanisms, because a feasibility study that hides its boundary conditions has not established feasibility. Within those boundaries, the case is made: an audited, shielded, on-device SLM junction controller is possible today, at 0.6B, on hardware a signal cabinet could house — and the path from here runs through closed-loop objectives, synthesized shields, and a watchdog-supervised field pilot.

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Chapter A

Reproducibility artifacts

[TODO: Pins table (Foundry 0.8.119, ort-genai builder, MiKTeX); seed lists; prompt templates verbatim; example audit record + verification transcript; compile-chain pitfall table; hardware inventory (RTX 2060 6GB serving, RTX 4090 training, RTX 4060 reserve).]

Chapter B

Extended results tables

[TODO: Full factorial cell tables; per-seed closed-loop tables; complete 7-row offline table with latency percentiles.]

Chapter C

UK-law fault-grounding knowledge base

[TODO: Port 01-research/uk-traffic-law.md structure: statutes, cases, the conditional EV exemption, the conflicting-green misfeasance carve-out.]